Computer Vision – Filters

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Filters

1. Given the following single row of an image with 9 gray levels (0-8):
   1, 0, 2, 2, 2, 8, 2, 3, 6, 6, 6
   Filter this image row

   filter mask needs to be flipped (convolution)

   a. with filter mask 1/3 [1 1 1] — x, 1, 1, 2, 4, 4, 4, 3, 5, 6, x — averaging filter
   b. with filter mask [1 0 -1] — x, 1, 2, 0, 6, 0, -5, 4, 3, 0, x
   c. with a 1x3 median filter — x, 1, 2, 2, 2, 2, 3, 3, 6, 6, x

2. The following table shows a part of a gray value image. Filter the 4 pixels in the middle (printed in bold) and enter the results in the empty tables below.

1	1	4	4
1	**1**	**4**	4
1	**1**	**4**	4
1	1	4	4

a. smoothened
b. tells us that brightness goes up and then back down (0 6 0 -5)
c. even more smoothened than a. but preserves edges

   a. use a 3x3 mean filter

	2	3	
	2	3	

would not copy down the non-filtered values, just keep them empty

   b. use a 3x3 median

	1	4	
	1	4	

3. The following table shows a part of a gray value image. Filter the 24 pixels in the middle (printed in bold) and enter the results in the empty tables below.

1	1	1	1	4	4	4	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	**1**	**1**	**1**	**4**	**4**	**4**	4
1	1	1	1	4	4	4	4

x direction (flipped)

-1 0 1
-2 0 2
-1 0 1

y direction (flipped)

-1 -2 -1
0 0 0
1 2 1

a. Use a 3x3 Sobel (derivation in x-direction)

	0	0	12	12	0	0	
	0	0	12	12	0	0	
	0	0	12	12	0	0	
	0	0	12	12	0	0	

b. use a 3x3 Sobel (derivation in y-direction)

	0	0	0	0	0	0	
	0	0	0	0	0	0	
	0	0	0	0	0	0	
	0	0	0	0	0	0	

Python/OpenCV - Task 1

Familiarize yourself with OpenCV. In particular, try out the following functions with a test image of your choice:

- Reading and writing images: `imread`, `imwrite`, `imshow`
- Gray conversion: `BGR2GRAY`
- Threshold value, binarization: `threshold`, `THRESH_OTSU`
- Histogram: `hist`

Python/OpenCV - Task 2

Load a test image of your choice and convert it to grayscale. Then write Python programs (or functions) with which you can perform the following tasks.

a) Histogram equalization with `equalizehist`
b) Filters: `GaussianBlur`, `filter2D`, `Sobel`, `Laplacian`.
c) Comparison of a linear low-pass filter (Gaussian, mean value) with a median filter (`medianBlur`)
d) First apply a linear low-pass filter to the image and then a high-pass filter. What happens if you enlarge the window of the low-pass filter (e.g. to 5x5, 7x7, 15x15)? Compare the result with the result you get when you replace the linear low-pass filter with a median.
e) Erosion, dilatation: `erode`, `dilate`
First, write "opening" and "closing" functions based on erosion/dilation. Also compare with the `morphologyEx` function. Special test images are available for this in the Learning Campus.